

Task 1: Data Modeling & Cleaning

- Load and transform datasets in Power Query
- Handle duplicates, missing values, and ensure correct data types

7 COLUMNS, 300 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:36 AM

● Create calculated columns:

○ Membership Duration = Today – Membership Since

Table: Customer_Demographics (300 rows) Column: Membership_Duration (11 distinct values)

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○ Extract Transaction Year, Transaction Month

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NameTransaction_YearFormatWhole numberSummarizationSumData typeWhole number

Formatting\$ % 0

PropertiesData categoryUncategorized

Sort by columnSortData groupsData groupsManage relationshipsRelationshipsNew columnCalculations

StructureFormattingProperties

1 Transaction_Year = YEAR(Customer_Transactions[Transaction_Date])

Customer_ID	Transaction_Year	Transaction_Month	Transaction_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Applied
CUST0047	2022	10	TXN00005	STORE048	Footwear	Friday, October 28, 2022	181.87	Yes
CUST0099	2023	1	TXN00007	STORE048	Accessories	Wednesday, January 11, 2023	63.7	No
CUST0159	2022	9	TXN00043	STORE048	Apparel	Wednesday, September 28, 2022	291.05	No
CUST0120	2022	9	TXN00018	STORE007	Apparel	Sunday, September 25, 2022	295.55	Yes
CUST0060	2025	8	TXN00027	STORE011	Accessories	Sunday, August 10, 2025	277.07	No
CUST0152	2024	1	TXN00015	STORE019	Accessories	Monday, January 22, 2024	80.96	No
CUST0074	2023	11	TXN00016	STORE030	Accessories	Wednesday, November 22, 2023	386.82	No
CUST0034	2024	6	TXN00022	STORE030	Footwear	Sunday, June 30, 2024	496.7	Yes
CUST0252	2024	7	TXN00034	STORE030	Footwear	Monday, July 15, 2024	263.96	No
CUST0237	2025	7	TXN00017	STORE046	Footwear	Tuesday, July 1, 2025	122.56	No
CUST0059	2024	12	TXN00021	STORE046	Accessories	Tuesday, December 3, 2024	219.19	Yes
CUST0155	2024	9	TXN00051	STORE022	Accessories	Wednesday, September 4, 2024	425.52	Yes
CUST0282	2022	12	TXN00057	STORE048	Apparel	Friday, December 30, 2022	259.53	Yes
CUST0295	2025	6	TXN00061	STORE035	Accessories	Saturday, June 21, 2025	228.53	No
CUST0164	2023	11	TXN00063	STORE011	Apparel	Tuesday, November 21, 2023	327.44	No
CUST0195	2025	2	TXN00066	STORE007	Footwear	Saturday, February 8, 2025	199.66	Yes
CUST0152	2025	6	TXN00068	STORE025	Apparel	Monday, June 16, 2025	192.37	No
CUST0053	2023	4	TXN00069	STORE019	Apparel	Tuesday, April 25, 2023	301.85	Yes
CUST0192	2022	10	TXN00073	STORE022	Footwear	Sunday, October 9, 2022	392.03	Yes
CUST0145	2022	9	TXN00074	STORE030	Apparel	Wednesday, September 14, 2022	195.21	Yes
CUST0267	2024	5	TXN00075	STORE007	Accessories	Monday, May 6, 2024	337.21	No
CUST0018	2025	2	TXN00076	STORE022	Footwear	Tuesday, February 11, 2025	342.57	No

Table: Customer_Transactions (1,000 rows) Column: Transaction_Year (4 distinct values)

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Formatting\$ % 0

PropertiesData categoryUncategorized

Sort by columnSortData groupsData groupsManage relationshipsRelationshipsNew columnCalculations

StructureFormattingProperties

1 Transaction_Month = MONTH(Customer_Transactions[Transaction_Date])

Customer_ID	Transaction_Year	Transaction_Month	Transaction_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Applied
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CUST0018	2025	2	TXN00076	STORE022	Footwear	Tuesday, February 11, 2025	342.57	No

Table: Customer_Transactions (1,000 rows) Column: Transaction_Month (12 distinct values)

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Final Conclusion

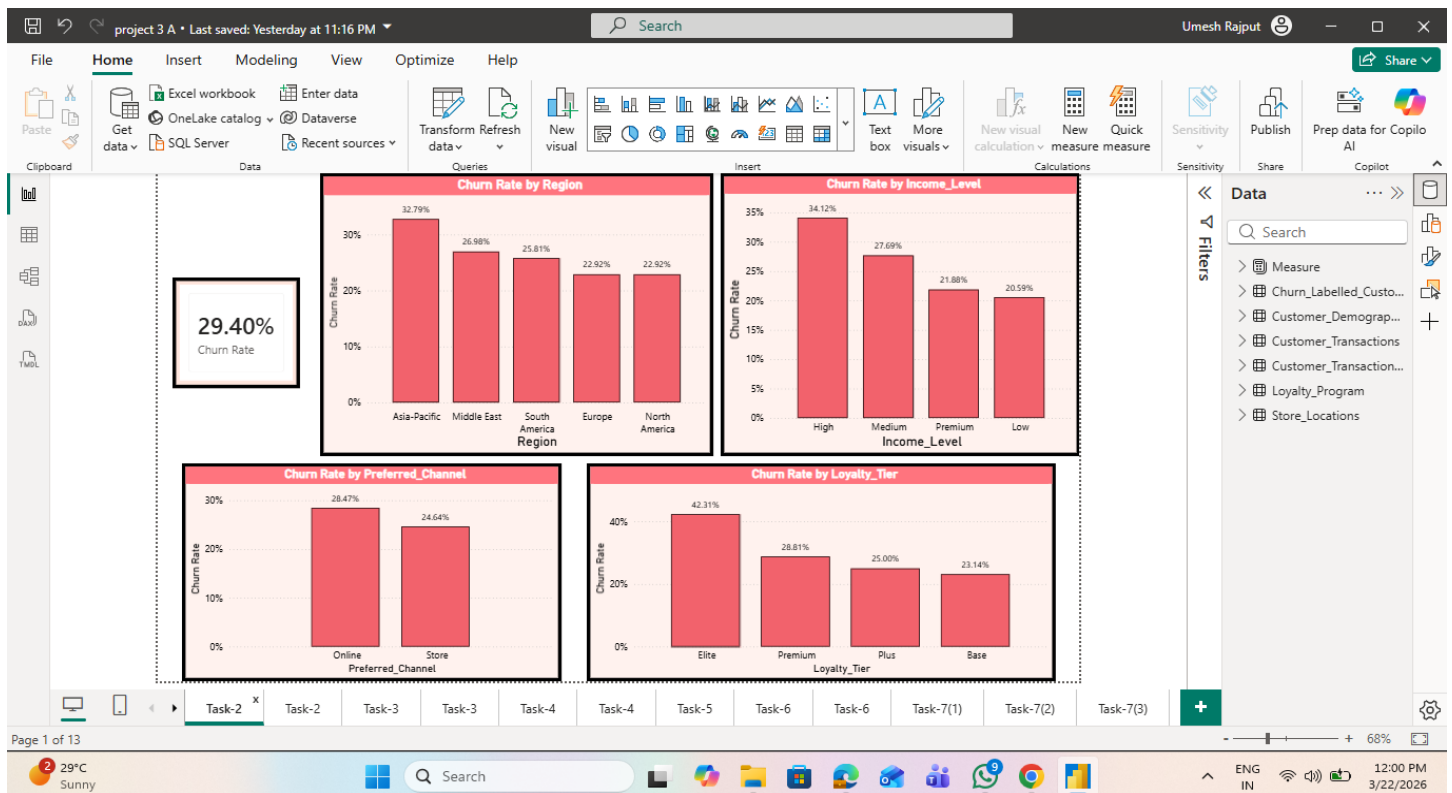
The dataset was successfully cleaned and transformed using Power Query by removing duplicates, handling missing values, and assigning correct data types. Key calculated columns like Membership Duration and transaction date fields were created, ensuring the data is well-structured and ready for accurate analysis and reporting.

Task 2: Churn & Retention Metrics

● Create Churn Rate KPI = (Churned Customers / Total Customers) * 100

● Visualize churn rate by:

○ Region ○ Income Group ○ Channel (Store/Online) ○ Loyalty Tier

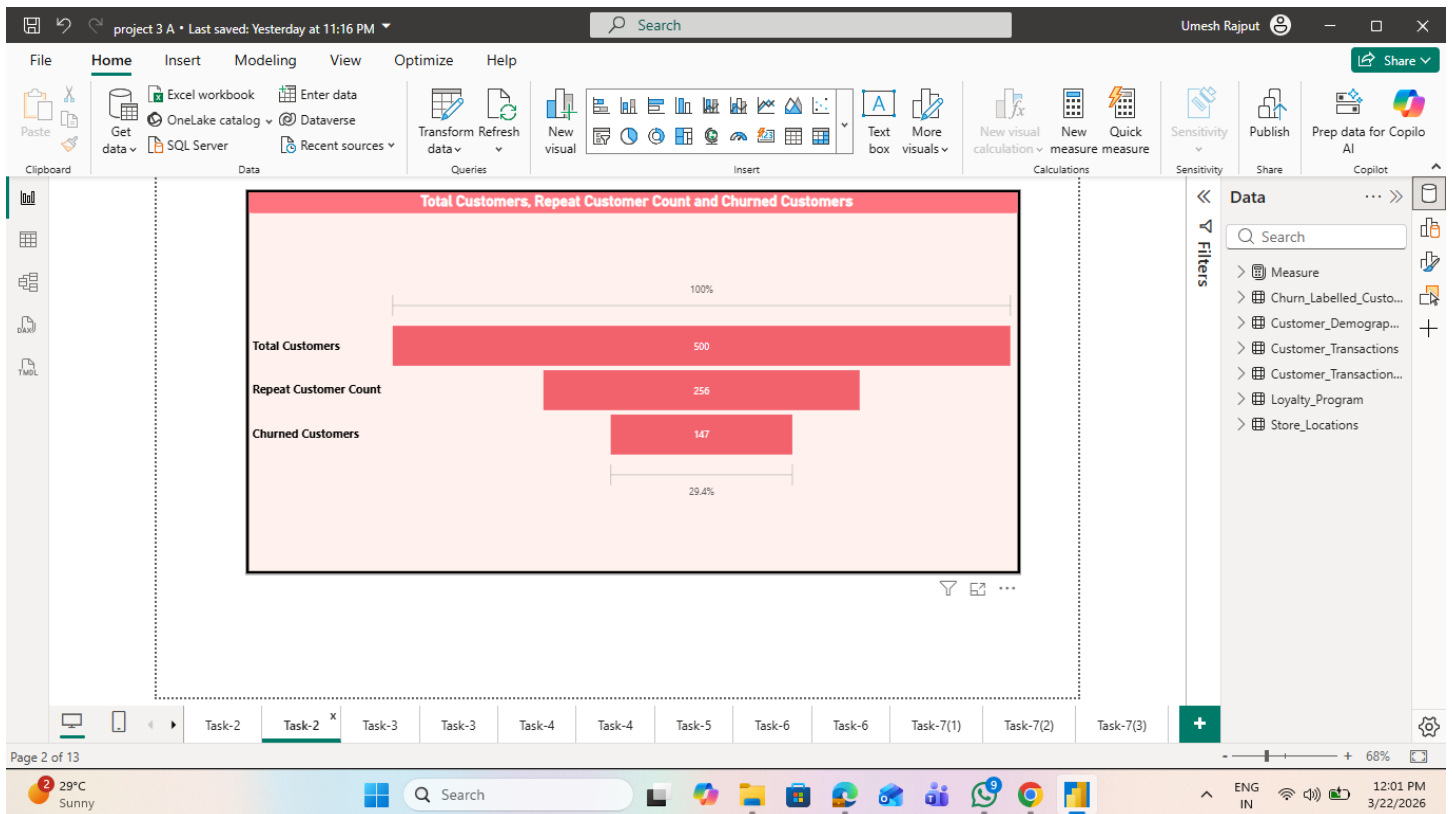


● Key Measures

1 – Churn Rate

✗ ✓ 1 Churn Rate = `DIVIDE([Churned Customers],[Total Customers])`

● Funnel Chart: Total Customers → Repeat Customers → Churned



● Key Measures

1 – Churned Customers

✖

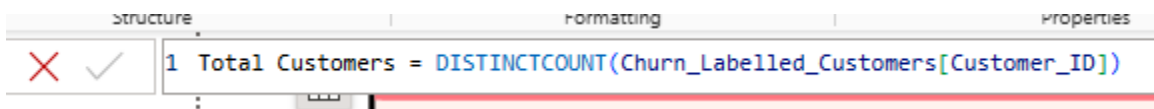
✔

1 Churned Customers = SUM(Churn_Labelled_Customers[Churn_Flag])

2 – Repeat Customer Count

Structure	Formatting	Properties	Calculations
✖ ✔ 1 Repeat Customer Count = 2 CALCULATE(3 DISTINCTCOUNT('Customer_Transactions'[Customer_ID]), 4 FILTER(5 VALUES('Customer_Transactions'[Customer_ID]), 6 CALCULATE(COUNT('Customer_Transactions'[Transaction_ID])) > 1 7) 8)			

3 - Total Customers



Final Conclusion

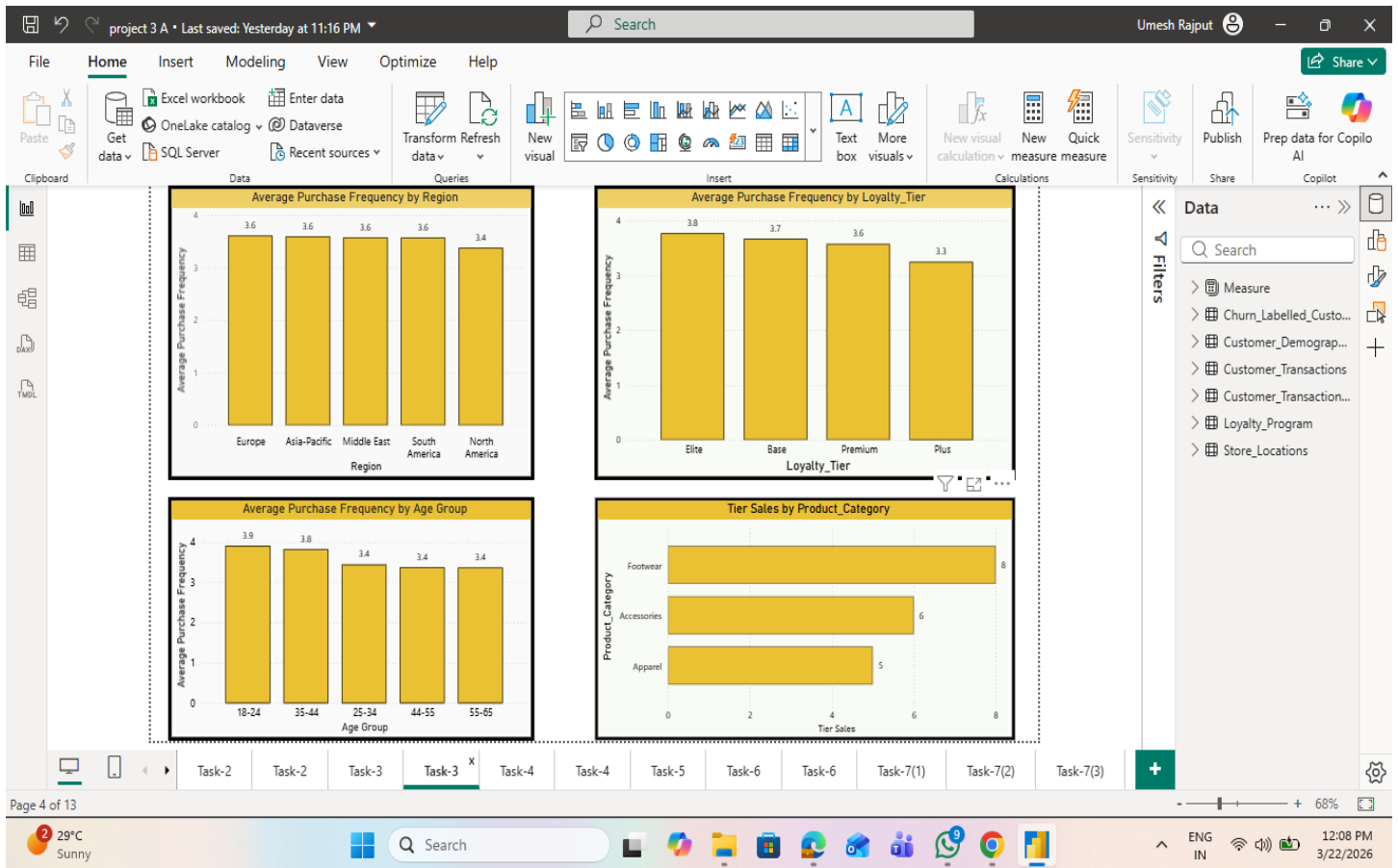
The overall churn rate is 29.40%, with higher churn in Asia-Pacific, high-income customers, and the Elite loyalty tier. The funnel shows a drop from 500 total to 256 repeat customers, indicating retention gaps. Overall, stronger strategies are needed to retain high-value customers.

Task 3: Repeat Purchase Analysis

- Segment customers
 - Low-Tier: 0–3 purchases
 - Mid-Tier: 4–8 purchases
 - High-Tier: 9+ purchases

Customer_ID	Count of Purchases	Segment customer
CUST0001	7	Mid-Tier
CUST0002	2	Low-Tier
CUST0003	2	Low-Tier
CUST0004	7	Mid-Tier
CUST0005	5	Mid-Tier
CUST0007	4	Mid-Tier
CUST0008	3	Low-Tier
CUST0009	5	Mid-Tier
CUST0010	2	Low-Tier
CUST0011	7	Mid-Tier
CUST0012	5	Mid-Tier
CUST0014	6	Mid-Tier
CUST0015	1	Low-Tier
CUST0016	5	Mid-Tier
CUST0017	9	High-Tier
CUST0018	5	Mid-Tier
CUST0019	4	Mid-Tier
CUST0020	7	Mid-Tier
CUST0021	1	Low-Tier
CUST0022	5	Mid-Tier
CUST0023	5	Mid-Tier
CUST0024	5	Mid-Tier
CUST0025	5	Mid-Tier
CUST0026	3	Low-Tier
CUST0027	5	Mid-Tier
CUST0028	5	Mid-Tier
CUST0029	1	Low-Tier
CUST0030	2	Low-Tier
CUST0031	5	Mid-Tier
CUST0032	2	Low-Tier
CUST0033	5	Mid-Tier
CUST0034	4	Mid-Tier
CUST0035	3	Low-Tier

- Compare avg. purchase frequency by Region, Age Group, Loyalty Tier
- Identify most purchased product categories by loyal customers



● Key Measures

1 – Average Purchase Frequency

✗ ✓ 1 Average Purchase Frequency = AVERAGE('Customer_Transactions (2)'[Count of Purchases])

2 – Tier Sales

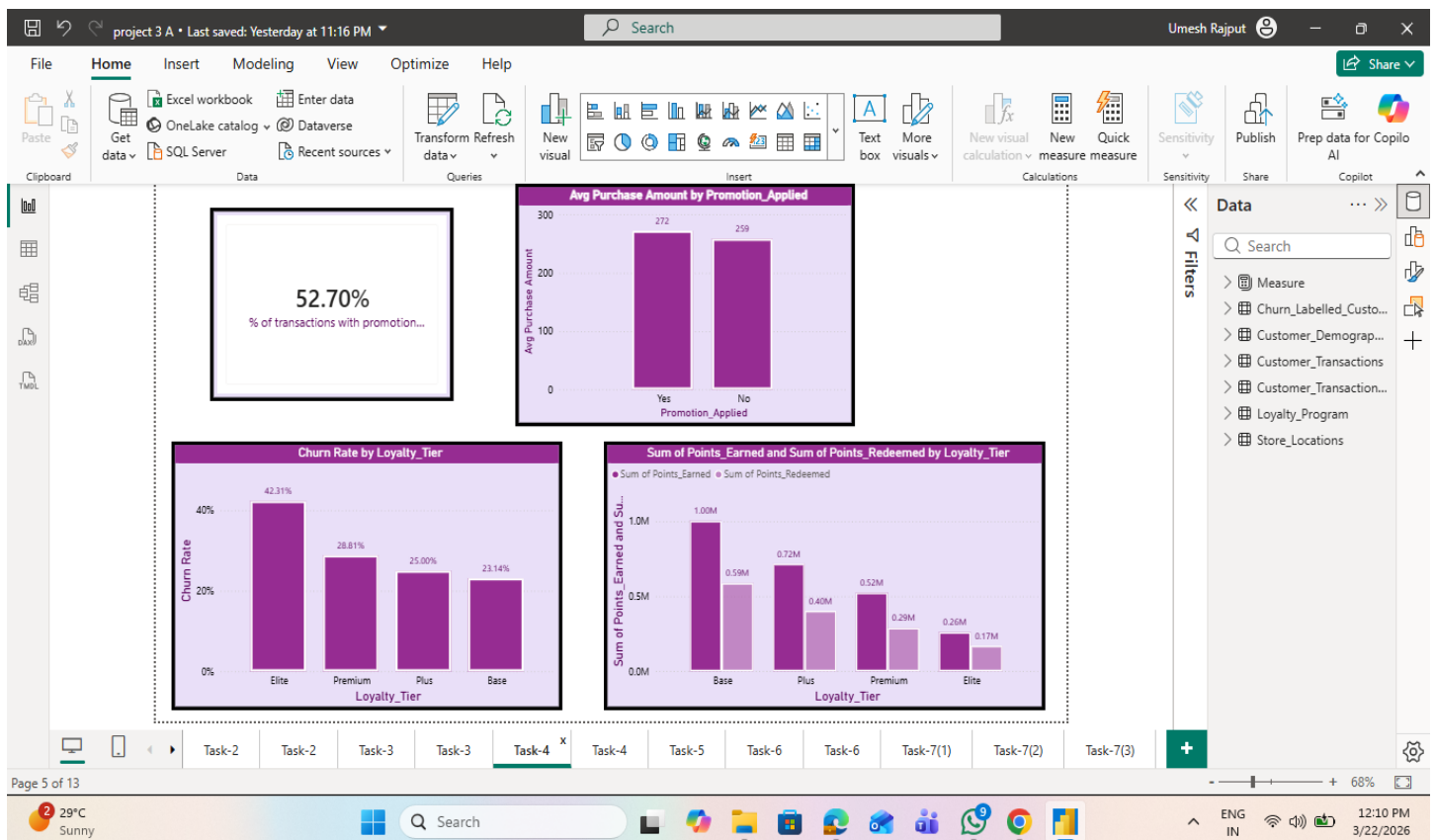
✗ ✓ 1 Tier Sales = CALCULATE(COUNT(Customer_Transactions[Product_Category]),'Customer_Transactions (2)'[Segment customer] = "High-Tier")

Final Conclusion

Most customers fall in the Mid-Tier segment (4–8 purchases), indicating moderate repeat behavior. Average purchase frequency is fairly consistent across regions (~3.4–3.6), with slightly higher engagement in the 18–24 age group and Elite loyalty tier. Among product categories, Footwear is the most purchased by loyal customers, followed by Accessories and Apparel.

Task 4: Promotion & Loyalty Impact

- % of transactions with promotion applied
- Compare avg. purchase amount with vs without promotions
- Churn rate across loyalty tiers
- Points Earned vs Redeemed by Tier (clustered column chart)



● Key Measures

1 - % of transactions with promotion applied


```
1 % of transactions with promotion applied = DIVIDE([Transactions with promotion applied],[Total Transactions],0)
```

2 – Avg Purchase Amount

```
1 Avg Purchase Amount = AVERAGE(Customer_Transactions[Amount])
```

3 – Churn Rate

```
1 Churn Rate = DIVIDE([Churned Customers],[Total Customers])
```

● Recommendations to improve redemption & retention

Recommendations

Since only 52.70% of transactions use promotions and purchases with promotions are higher, expanding promotional campaigns can help increase overall sales. The Elite tier shows the highest churn (~42%), indicating a need for stronger retention strategies such as exclusive rewards and personalized offers. Additionally, points earned exceed points redeemed across all tiers, especially in the Base tier, showing low utilization of rewards. To improve this, the redemption process should be simplified, and businesses should offer limited-time rewards, bonus point campaigns, and send reminders for unused or expiring points to enhance customer retention and engagement.

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Final Conclusion

The analysis shows that promotions positively influence customer spending, as transactions with promotions have higher purchase value. However, the high churn rate in higher loyalty tiers, especially Elite, indicates weak retention. Additionally, points earned are higher than points redeemed across all tiers, showing low utilization of rewards. Overall, better promotional strategies, improved redemption systems, and targeted loyalty benefits are needed to enhance customer retention and engagement.

Task 5: Store & Channel Performance vs Retention

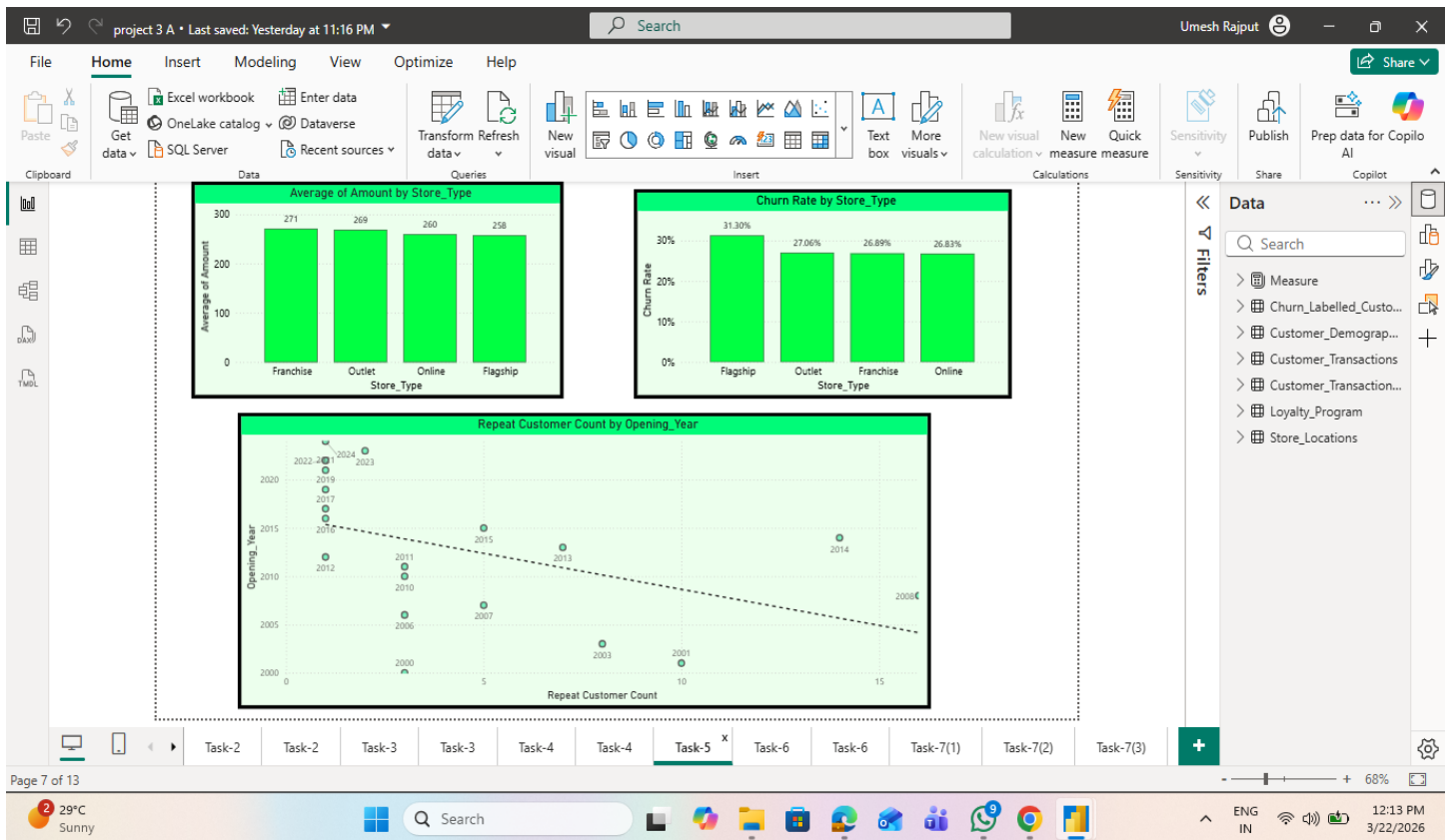
● Merge store data with transactions

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with the following columns: **Promotion_Applied**, **Store_Locations.Store_Type**, **Store_Locations.Region**, and **Store_Locations.Opening_Year**. The table contains 23 rows of data. The right sidebar shows the **Query Settings** pane, which includes the **PROPERTIES** section (Name: Customer_Transactions) and the **APPLIED STEPS** section, where **Expanded Store_Locations** is selected.

	Promotion_Applied	Store_Locations.Store_Type	Store_Locations.Region	Store_Locations.Opening_Year
1	Yes	Outlet	Asia-Pacific	2013
2	Yes	Outlet	Asia-Pacific	2013
3	Yes	Online	South America	2017
4	Yes	Flagship	South America	2013
5	Yes	Flagship	South America	2013
6	Yes	Franchise	Asia-Pacific	2011
7	Yes	Franchise	South America	2014
8	Yes	Outlet	Middle East	2000
9	No	Franchise	Europe	2013
10	Yes	Franchise	Europe	2013
11	Yes	Outlet	North America	2003
12	No	Outlet	North America	2003
13	No	Outlet	North America	2003
14	Yes	Online	Europe	2014
15	No	Franchise	Europe	2001
16	Yes	Franchise	Europe	2001
17	Yes	Outlet	Europe	2015
18	Yes	Online	North America	2017
19	No	Online	Europe	2010
20	Yes	Online	Europe	2010
21	No	Online	Europe	2010
22	Yes	Online	Europe	2006
23	No	Online	Europe	2006

● Visualize

- Avg. transaction amount by Store Type
- Churn rate by store type
- Correlation between store opening year & retention



● Key Measures

1 – Churn Rate

$$1 \text{ Churn Rate} = \text{DIVIDE}([\text{Churned Customers}], [\text{Total Customers}])$$

2 – Repeat Customer Count

```

1 Repeat Customer Count =
2 CALCULATE(
3     DISTINCTCOUNT('Customer_Transactions'[Customer_ID]),
4     FILTER(
5         VALUES('Customer_Transactions'[Customer_ID]),
6         CALCULATE(COUNT('Customer_Transactions'[Transaction_ID])) > 1
7     )
8 )

```

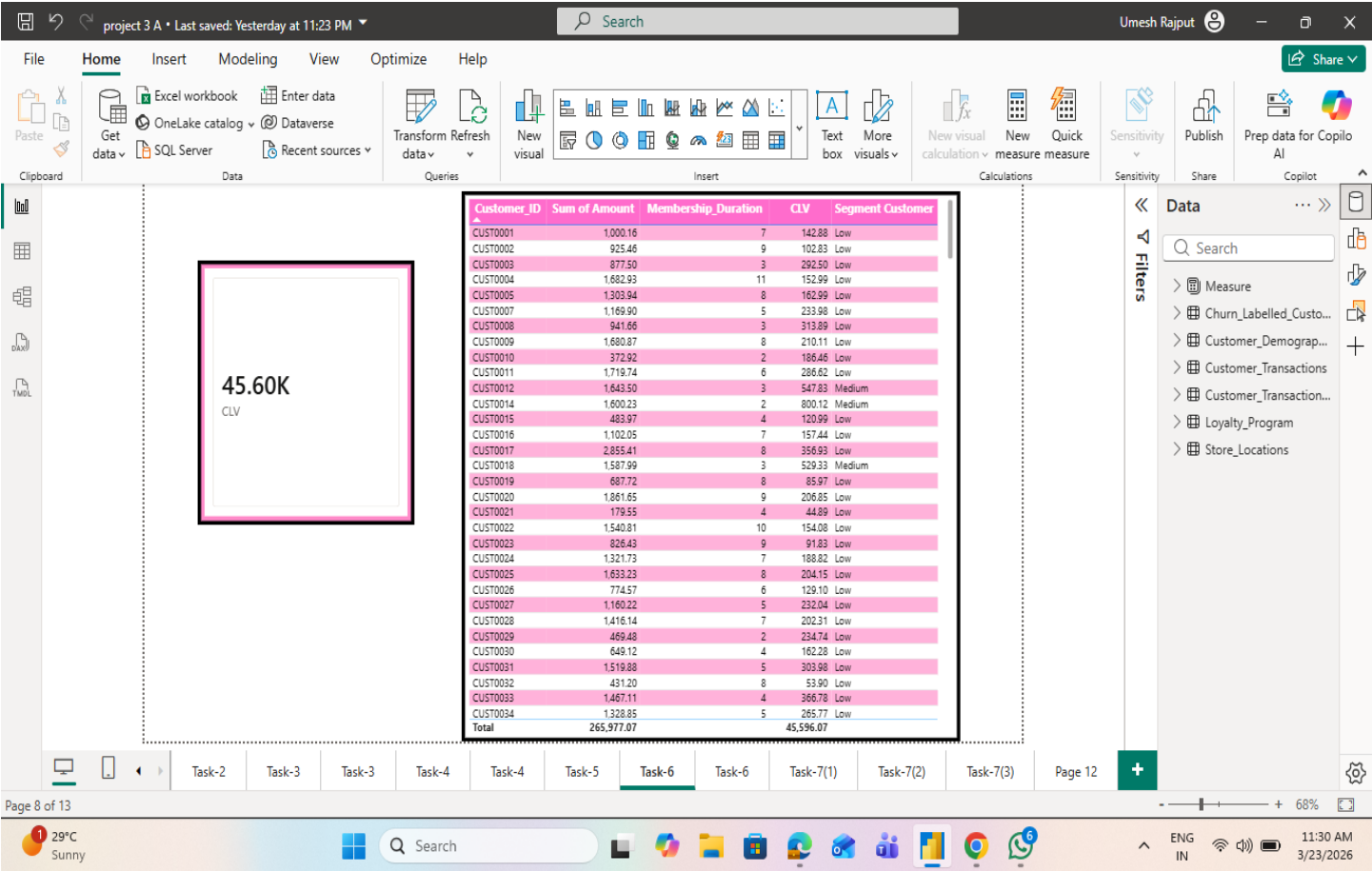
Final Conclusion

Store and transaction data were merged using Store-ID to analyze performance. The results show that Franchise and Outlet stores generate the highest average transaction value, while Flagship stores have the lowest value and highest churn rate (31.30%), indicating poor retention. Online stores have the lowest churn, showing better customer engagement.

The correlation analysis indicates that older stores have higher customer retention than newly opened stores. Thus, while revenue performance is strong in some store types, retention issues exist in Flagship and new stores, requiring improvement in customer experience and loyalty strategies.

Task 6: Customer Lifetime Value (CLV) Analysis

- $CLV = \text{Total Amount Spent} / \text{Membership Duration (Years)}$
- Segment customers into Low, Medium, High CLV



● Key Measures

1 – CLV

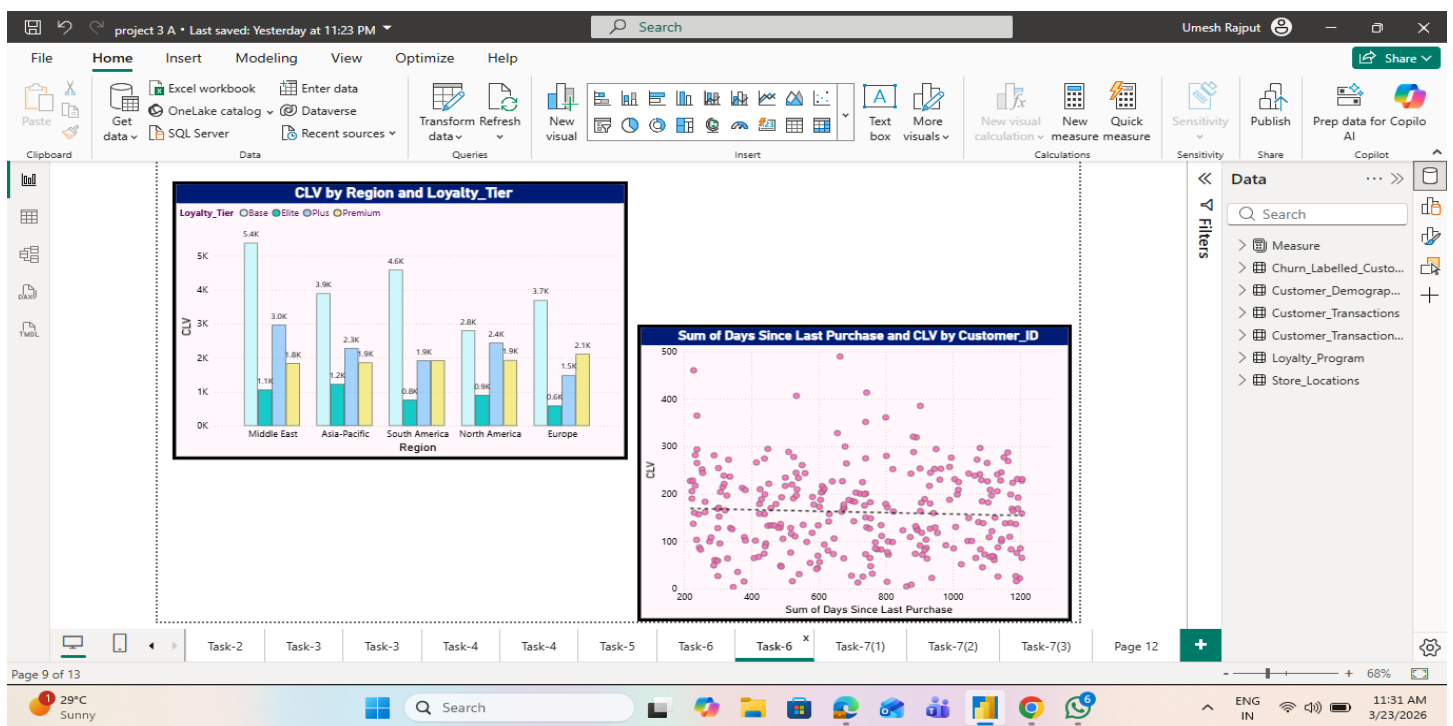
```
1 CLV = DIVIDE(SUM(Customer_Transactions[Amount]),AVERAGE(Customer_Demographics[Membership_Duration]),0)
```

2 – Segment Customer

```
1 Segment Customer =  
2 IF(  
3     [CLV] <= 500, "Low",  
4     IF(  
5         [CLV] <= 1500, "Medium",  
6         "High"  
7     )  
8 )
```

● Visualize:

- CLV vs Days Since Last Purchase
- CLV by Loyalty Tier & Region



Final Conclusion

Customer Lifetime Value (CLV) analysis shows that although the total CLV is around 45.6K, most customers fall in the Low CLV segment (below ~300–400), with very few in Medium and almost none in High, indicating low overall customer value. The relationship between CLV and days since last purchase is weak, which suggests inconsistent customer engagement. Additionally, Elite tier customers generate

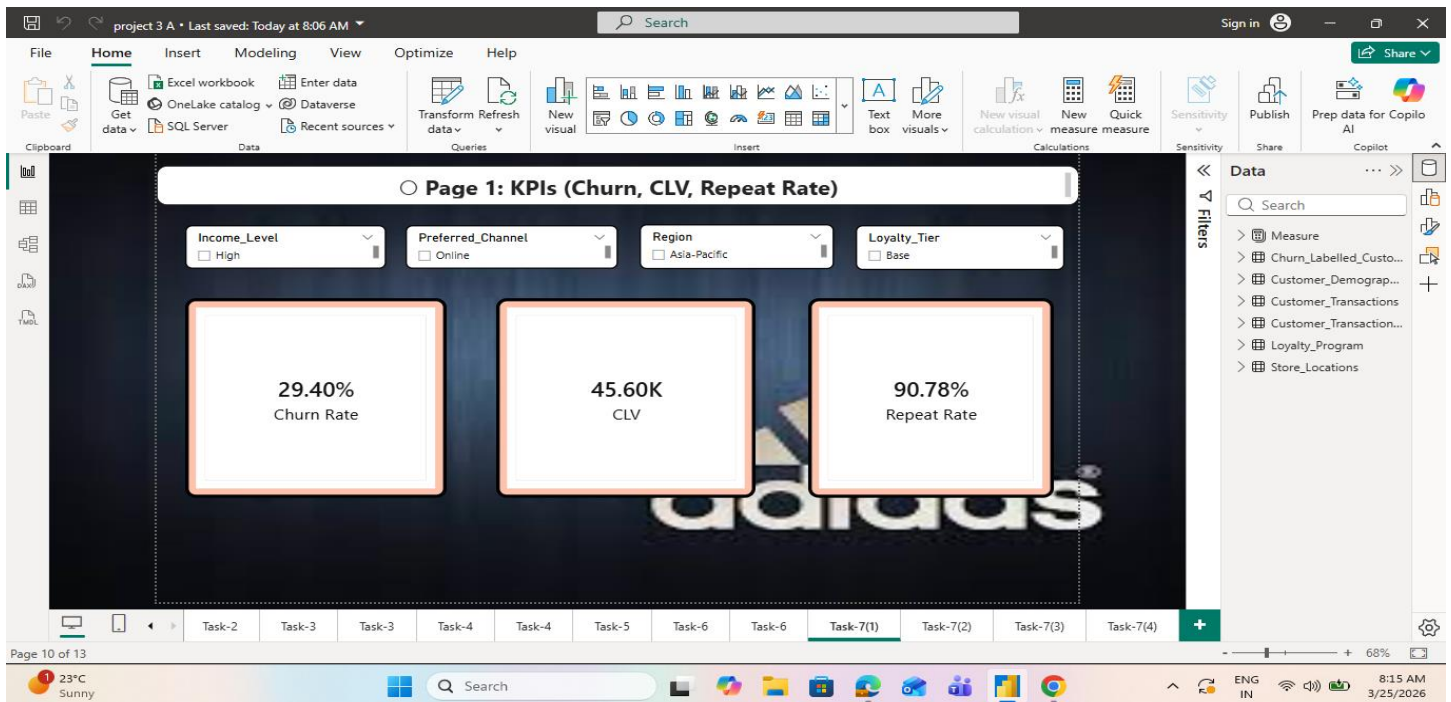
the highest CLV across all regions (e.g., highest in Middle East and South America), while Base tier contributes the least. Overall, the business is mainly dependent on low-value customers, and improving engagement and retention is essential to increase CLV and move customers into higher-value segments.

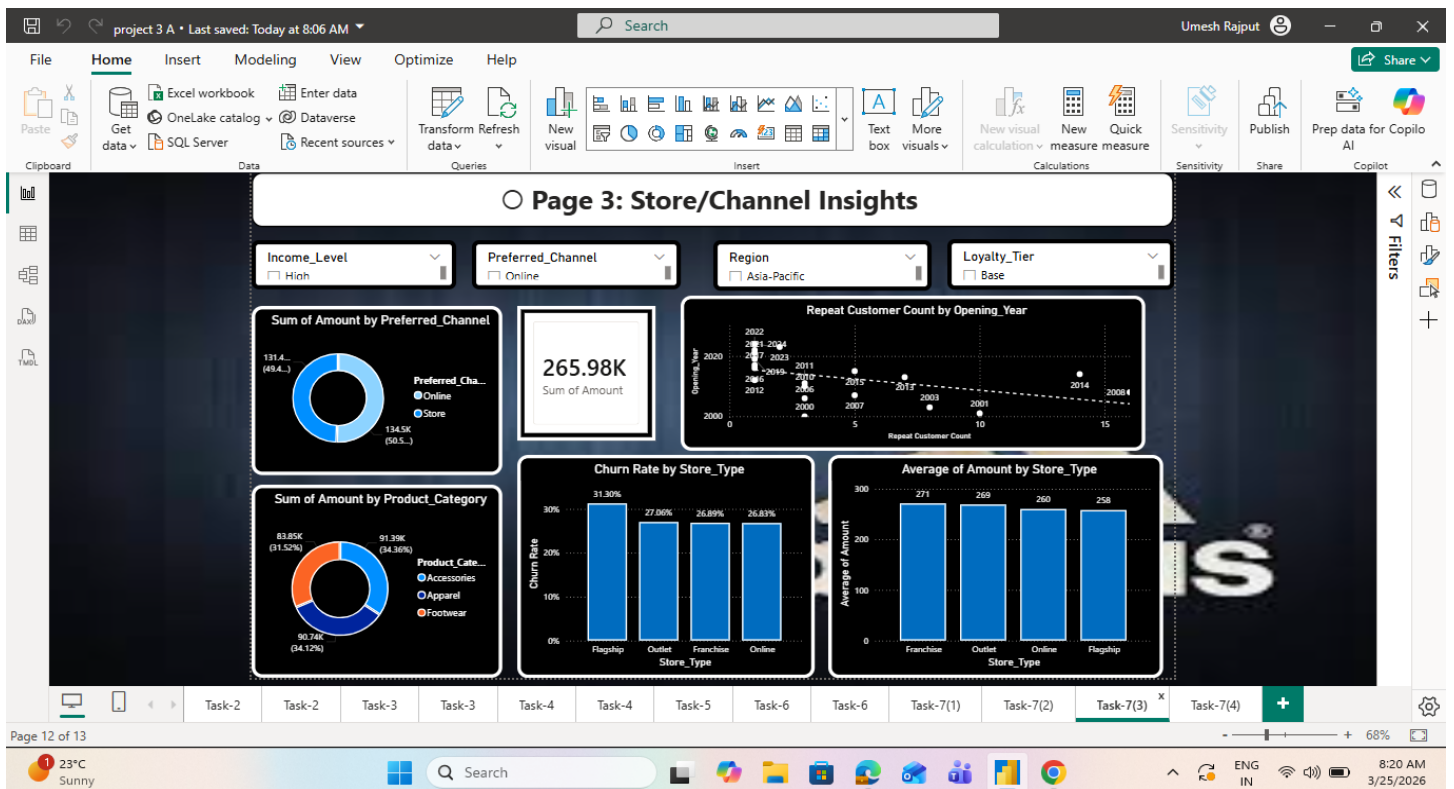
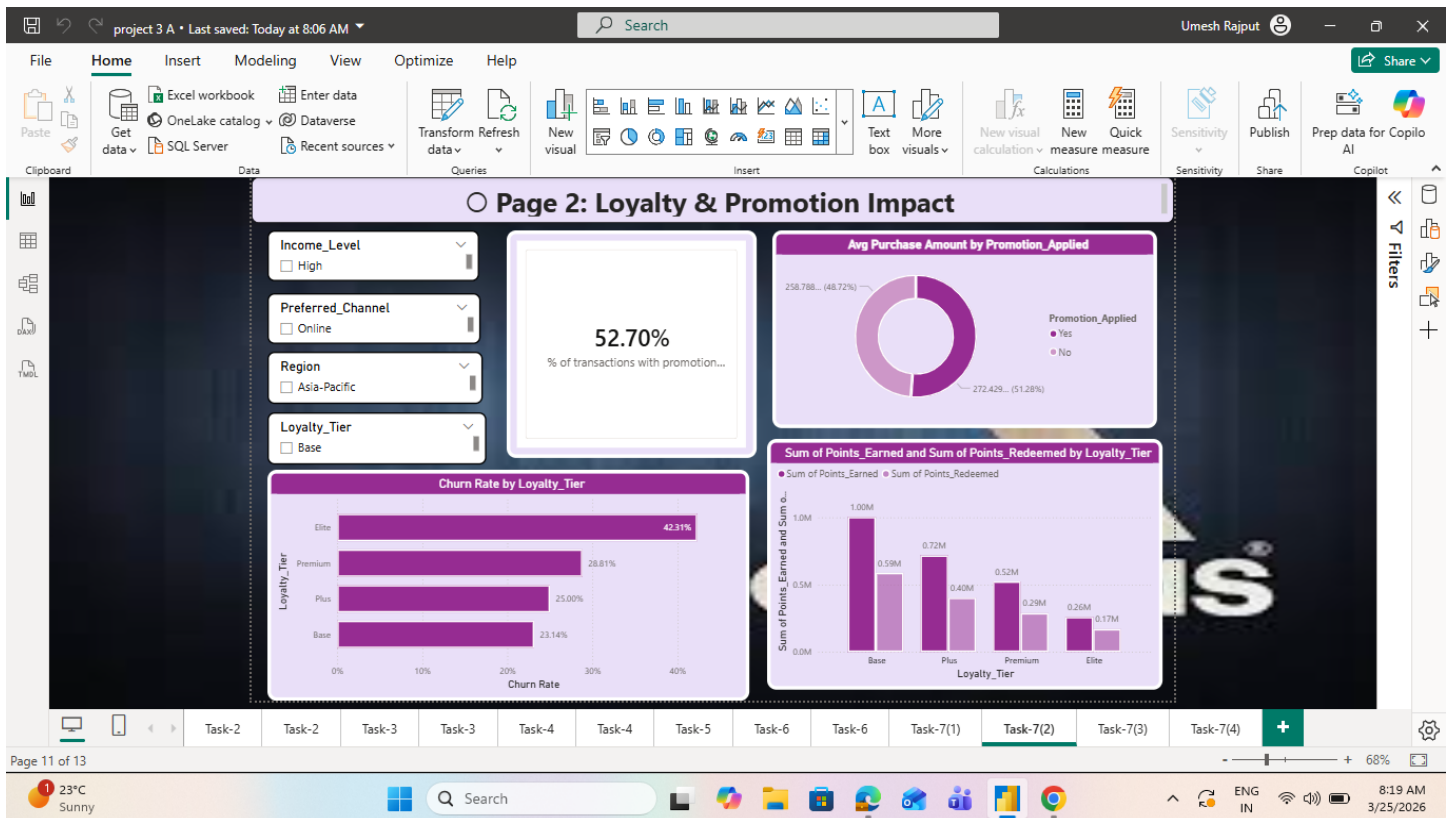
Task 7: Final Dashboard & Executive Summary

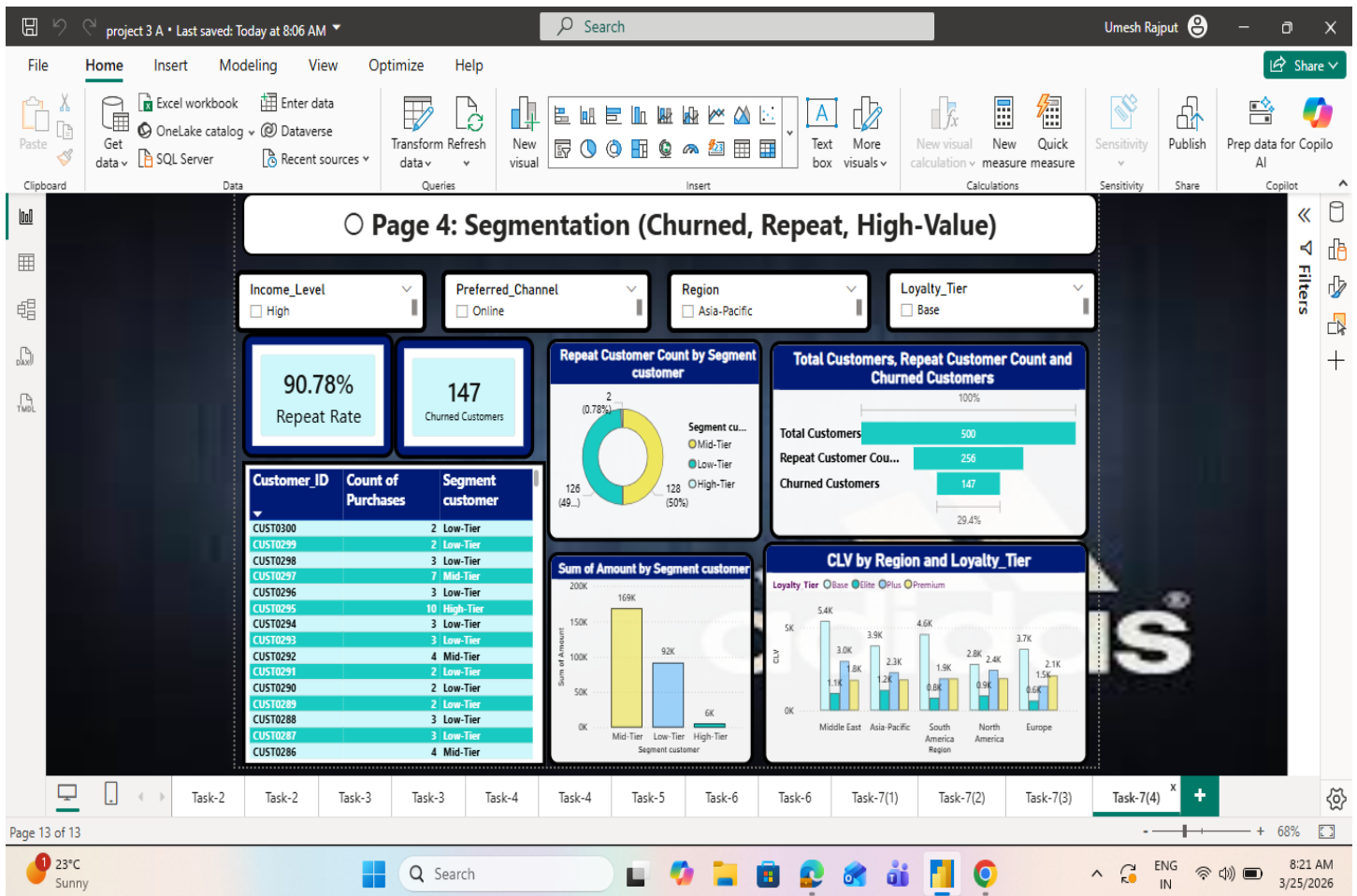
● Multi-page Power BI Report

- Page 1: KPIs (Churn, CLV, Repeat Rate)
- Page 2: Loyalty & Promotion Impact
- Page 3: Store/Channel Insights
- Page 4: Segmentation (Churned, Repeat, High-Value)

● Slicers: Region, Channel, Income, Loyalty Tier







● Top 3 recommendations for Adidas

- Which customers to prioritize for retention?
- Which channels are underperforming?
- How to strengthen loyalty program engagement?

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Top 3 recommendations for Adidas

○ Which customers to prioritize for retention?

Adidas should focus on Elite and Premium customers, as they have the highest churn rate (~42%) despite high CLV. Additionally, high-income customers should also be targeted since they show higher churn but contribute more revenue.

○ Which channels are underperforming?

The Online channel is underperforming as it has the highest churn rate (~28%) compared to store purchases. Also, Flagship stores show relatively higher churn, indicating a need for better customer experience.

○ How to strengthen loyalty program engagement?

Adidas should improve the loyalty program by
Offering better and faster reward redemption
Providing personalized offers
Introducing exclusive benefits for Elite members

Data

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Loyalty_Program

Store_Locations

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